



Instructor: Date:

Learner: Regional Branch:

Program: Center:

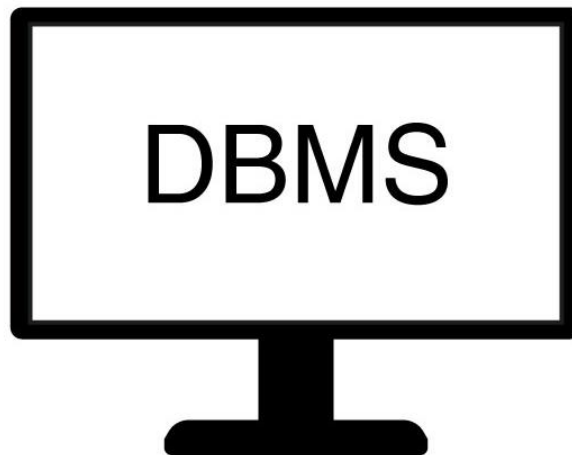
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Let's Start!

Organizing Information



Learning Outcome:

Exchange information about relational database management system.

Ask your partner:

- How do you organize the information in your computer? Why do you use that strategy?
- Do you think the strategy you use is efficient? Why?

Can Do Statements:

- I can say what a database management system is.
- I can list the characteristics a modern DBMS has.
- I can talk about the characteristics of modern DBMS.





Let's Explore!

Carlos helps Lilly understand key aspects about DBMS.



Instructions: Read the text below.

- Lilly:** Hi, Carlos! Can you come here? I heard that we need to use a DBMS in our next project, but I don't remember very well what that means.
- Carlos:** Don't worry, I can help you with that. Let's go step by step.
- Lilly:** All right, I am ready.
- Carlos:** So am I. A database is the collection of data, which we can also refer to as facts and figures; plus, we process that data to get information. Now, when we talk about the database management system, or DBMS, we have to say that we refer to the way we store that data. When we use DBMS, we organize data in a way it can be found and manipulated easily; furthermore, we can also produce more information from it.
- Lilly:** Then, I guess that when we use DBMS, we don't have to use file formats as we do with databases.
- Carlos:** Neither do I. In fact, there are some special characteristics that any modern DBMS must have in order to work properly. For example, they must use and have real-world entities, relation-based entities, isolation of data, no redundancy, consistency, query language, ACID properties, multiple views, and security.
- Lilly:** Wow, can you explain some of those characteristics?
- Carlos:** Sure. Well, as you have to design the architecture of the DBMS, you need to base it on real-world entities; plus, the tables in which the information will be organized have to follow relation-based entities. Furthermore, DBMS stores data about data, which we know as metadata, and it follows certain rules to avoid redundancy and ensure consistency. Additionally, we can use query language to perform all sort of commands to retrieve, manipulate, or delete information, among other actions; besides, we can also use the ACID properties to perform similar transactions. Finally, as the database may have different users, it will show different views for each one of them; and this actually increases security within the database.
- Lilly:** Thank you! I feel like I am an expert now.
- Carlos:** So do I. It's complicated at the beginning, but it's worth learning about it.



Let's Practice!

Let's review what happened between Lilly and Carlos.

Instructions: Use the words from the Word Bank to complete each sentence.

Word Bank

explains / knows / uses / teaches / explains

1. Carlos _____ a DBMS to store data in a non-traditional way.
2. Carlos wants Lilly to not get confused; that is why he _____ the difference between a database and a DBMS.
3. Carlos _____ how a DBMS works in a system, we can say he is an expert!
4. Carlos _____ Lilly that a DBMS is a big group of facts and figures that are organized according to specific criteria so they can be found easily.
5. Carlos _____ that we can produce more information out of the DBMS.



Let's Practice!

Carlos knows a lot about this DBMS.

Instructions: Select the correct answer.

1. Carlos understands all the important elements related to modern DBMS.
Hint: Check the basic components they must have.
 - a. So does Lilia. She knows that this system has a new battery, atomicity, consistency, isolation and durability.
 - b. So does Lilly. She can now say that this system must have future-world entities, data isolation, redundancy, query language, among others.
 - c. So does Lilly. Now she knows that DBMS must have real-world entities, ACID properties, multiple views, security, and data isolation, among others.
 - d. So does Mary. She now knows that those systems have a new code, ACID properties, consistency, isolation and strength.



Let's Practice!

2. Lilly is excited to learn about DBMS.
 - a. So is Carlos. Now both feel like they are experts.
 - b. Neither does Carlos. He wants to follow the traditional styles of data management.
 - c. Neither do I. I think they have to be updated this year.
 - d. So is Carlos. He knows that the database information can be stored in the servers.

3. Lilly believes that the DBMS does not use file formats at all.
Hint: The statement is a negative one.
 - a. Neither do Carlos. He has no clue at all.
 - b. So does Lilly. She knows it is a hard thing to do.
 - c. Neither does Lilly. However, she manages AXID properties very well.
 - d. Neither does Carlos. In fact, there are some specific components and characteristics we have to check for any modern DBMS.

4. Lilly understands that we can use query language to perform commands on the data, and that we can also use the ACID properties for such task.
Hint: Check the characteristics of DBMS.
 - a. So does Carlos as he gave Lilly that information.
 - b. Neither does Carlos. He hates technology.
 - c. Neither do Lilly and Carlos. They agree.
 - d. So did Carlos when he was a kid.

5. Lilly now thinks that as databases have different users and each one of them has a special access to the information, the security levels of the database increase.
Hint: The statement is affirmative.
 - a. So does Carlos. He is happy with the result.
 - b. So does Carlos. In fact, he is an expert on this topic.
 - c. So does Charles. He calls her to help her out.
 - d. Neither does Lilly. She has no one to help her out.



Extension Activity

Let's talk about the database management system.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for further information about database management system.
- Take notes about the information you read.
- Make a dialogue to explain the characteristics of a modern database management system.
- Use the simple present tense, and responses with 'so' and 'neither'.
- Share your dialogue with a partner.



Target Words

- **facts** (noun): elements that are real and exist
e.g. A database is the collection of data, which we can also refer to as facts and figures; plus, we process that data to get information.
- **figures** (noun): the use of numbers in a calculation
e.g. A database is the collection of data, which we can also refer to as facts and figures; plus, we process that data to get information.
- **sort** (noun): a particular series of objects; a class or group of elements
e.g. Additionally, we can use query language to perform all sort of commands to retrieve, manipulate, or delete information, among other actions.

Self-Assessment

Yes

No

- I search for information about database management system.
- I write a dialogue to explain the characteristics of a modern database management system.
- I use the simple present tense as well as responses with 'so' and 'neither'.
- I share my dialogue with a partner.

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- **retrieve** (verb): to recover or find something
e.g. Additionally, we can use query language to perform all sort of commands to retrieve, manipulate, or delete information, among other actions.
- **ACID properties** (noun): stands for Atomicity, Consistency, Isolation, Durability, which we use to perform different transactions to manipulate the data a database contains
e.g. Besides, we can also use the ACID properties to perform similar transactions.



Learning Points

Responses with So and Neither

You use **'so'** and **'neither'** to agree with what another person has previously said.

The responses given using **'so'** and **'neither'** show that we agree with someone without having to repeat what that person said. In fact, the statements can be positive or negative. However, both expressions differ in use.

'So' is used to show agreement with a previous positive statement. In fact, an equivalent expression for it would be "me too". The structure that we follow is:

So + verb 'to be' or auxiliary verb* + subject pronoun.

For example:

*Peter: I **understand** everything about SQL syntax.*

*Sally: **So do I.** (As Peter's present simple statement uses a verb different from the verb 'be', we use the auxiliary verb.)*

*Sally: I **am** an expert on computers.*

*Peter: **So is** Sally! (As Sally's statement uses the verb 'be', we use the same one here.)*

On the other hand, we use the expression **'neither'** to show agreement with negative statements. The structure that we use is the following:

Neither + verb 'to be' or auxiliary verb* + subject pronoun.

* The difference between using the verb **'be'** or the auxiliary depends on the verb that the previous statement uses.

For example:

*Peter: **I'm not** sure about SQL syntax. Sally: **Neither am I.***

(As Peter's statement uses the verb 'to be', the response that Sally gives has to use the same verb, too.)

*Sally: **I don't understand** anything about DBM.*

*Peter: **Neither do I.** (As Sally's present simple statement uses a verb different from the verb 'be', we use the auxiliary verb.)*

Examples

- Lilly: Then, I guess that when we use DBMS, we don't have to use file formats as we do with databases.*
Carlos: Neither do I.
- Lilly: Thank you! I feel like I am an expert now.*
Carlos: So do I.





Learning Points

Commands in Structured Query Language

You use SQL statements to write the action that needs to be performed on a database.

In SQL syntax there are some specific words called *commands* that are actually keywords that will help us manage or control the database; furthermore, they help us look for information in it.

The following chart displays a series of important commands that are frequently used in SQL query and that affect the databases to which they are applied:

COMMAND	USE
SELECT	Extracts information or data
DELETE	Removes information or data
CREATE TABLE	Generates a new table
UPDATE	Updates information or data
ALTER TABLE	Changes elements in a table
INSERT INTO	Puts new information or data into the database
CREATE DATABASE	Generates a new database
ALTER DATABASE	Changes elements in a database
DROP INDEX	Erases an index
CREATE INDEX	Generates a new index (search key)
DROP TABLE	Erases an index